PROTOCOL

AimGel Human NK Cell Activator

CATALOG ITEMS: ANK01-0005H, ANK01-0050H

FOR RESEARCH USE ONLY
STORE IN 2-8˚C

Description

AimGel hydrogel microparticles (HMP) are designed for robust activation of human NK cells using a feeder-free culturing approach. For any downstream processing (e.g. counting, flow cytometry) or earlier removal of HMP, please remove using the HMP lysis buffer provided in the kit.

Components

A suspension of hydrogel microparticles (HMP) made of chemically crosslinked dextran. HMPs sized 7-12 µm were coated with phospholipid bilayer, with membrane docked human NK cell activating signal panels. Each kit contains:

- 1 vial of AimGel Human NK Cell Activator in PBS at $4x10^7$ microparticles/mL, with 0.5% P/S
- 1 vial of HMP lysis buffer (100X), contains dextranase in PBS

Item	AimGel		HMP lysis buffer
ANK01-0005H	0.05 mL	HMP count: $2×10^6$	0.1 mL
ANK01-0050H	0.5 mL	HMP count: $2×10^7$	1.0 mL

Stability and Storage

- Shipped with ice, keep product refrigerated (2-8°C)
- Stable at 4°C for 12 months
- Contents are sterile in unopened tube
- Do not subject product to freezing, high temperatures (>25°C)

Other Required Materials

- ○ Cryopreserved or freshly isolated human $CD3^-CD56^+$ Cells
- ○ RPMI 1640 supplemented with 2 mM L-glutamine (or equivalent)
- ○ Heat inactivated human AB serum (hAB serum)
- ○ Penicillin/streptomycin
- ○ Recombinant human IL-2 (rIL-2)
- ○ Recombinant human IL-15 (IL-15)
- ○ 2-Mercaptoethanol
- ○ Cell culture vessels
- ○ Humidified CO_2 incubator or bioreactor

Activation Protocol

Medium preparation

Base media

RPMI 1640 supplemented with 2 mM L-glutamine (or equivalent), 1% P/S, 50 µM 2-mercaptoethanol, 25 mM HEPES, 1X non-essential amino acids.

Post-thaw recovery media

Base media with 10% heat-inactivated human AB serum, plus 500 U/mL IL-2 and 5 ng/mL IL-15. Add serum, cytokines and 2-mercaptoethanol freshly.

Expansion media

Base media with 5% heat-inactivated human AB serum, plus 500 U/mL IL-2.

Post-thaw recovery

For **cryopreserved NK cells**, we recommend an overnight recovery before activation to improve viability and shorten lag time. Prepare post-thaw recovery media freshly prior to use. Resuspend the NK cells in recovery media at $5×10^5$–$1×10^6$/mL and recover overnight (16–24 hours) to restore surface ligands lost during cryopreservation. Harvest the recovered cells and centrifuge at 500×g for 5 minutes to discard the recovery media. Resuspend the cell pellet in expansion media and count viable cells. Dilute to the appropriate concentration using expansion media.

HMP preparation

Resuspend HMP in the vial by vortexing for 30 seconds. Open HMP vial and HMP lysis buffer only in sterile environment to avoid contamination. Calculate the desired HMP seeding density per well and aliquot the HMP into each well accordingly. Rather than recommending a fixed bead-to-cell ratio, we provide a validated seeding density for reference. We also encourage experienced users to optimize seeding conditions based on their specific experimental requirements. The HMP seeding principle is to balance the interaction probability between HMP and target cells while leaving sufficient space for cells to grow.

Recommendations

Isolated $CD3^-CD56^+$ NK cell seeding density : $2.5×10^5$ cell per cm^2
HMP seeding density : $2.5×10^5$ HMP per cm^2

Plate	Area (cm^2)	NK/well	HMP/well	HMP Vol.
96-well	0.32	$8 ×10^4$	$8 ×10^4$	2.0 µL
48-well	0.95	$2.4×10^5$	$2.4×10^5$	5.9 µL
24-well	1.9	$4.8×10^5$	$4.8×10^5$	12 µL
6-well	9.5	$2.4×10^6$	$2.4×10^6$	60 µL
T25 flask	25	$6.3×10^6$	$6.3×10^6$	156 µL

Plate	Area (cm^2)	NK/well	HMP/well	HMP Vol.
T75 flask	75	1.9×10^7	1.9×10^7	469 µL

Cell seeding
Aliquot resuspended NK cells. Gently mix cells and HMP by pipetting up/down 3 to 5 times ensure HMP and cells are evenly distributed under microscope.

Initial activation
Incubate in a humidified 5% CO_2 incubator at 37°C. Monitor cell morphology and confluency, perform half media replenishment every 2-3 days with **Expansion media**.

HMP-cell clustering is a typical phenomena you should expect in the first week. **DO NOT disturb** HMP-cell interaction in the first 4 days to allow full activation. At Day 4-5, gently pipette cell & AimGel HMP suspensions up and down to break up clumps.

Determine cell density and replate cells to a larger culture vessel if concentration exceeds 2×10^6 cells/mL or top-up with fresh media to adjust cell concentration to 5×10^5 cells/mL.

Restimulation
NK cell growth typically slows after day 16 post-initial activation. Users may consider restimulation to promote continued cell expansion. Restimulating with AimGel HMP minimizes activation-induced exhaustion through its mild yet effective stimulation signal. We recommend using **1×10^5 HMP per cm^2** for restimulation.

HMP cleanup
Remove HMP by adding HMP lysis buffer (included in the kit) directly to the culture medium at 1× final concentration. After incubating at 37°C for 30-60 minutes, the HMP will be fully degraded.

AimGel Human NK Cell Activator ships in a non-hydrolysable formulation. A self-hydrolysing formulation, which degrades over approximately 20 days in serum-containing RPMI 1640, is available on request.

Important Note
- HMP lysis buffer contains dextranase, which degrades the HMP core. Do not use this buffer if you plan to restimulate cells, as residual enzyme will degrade newly added HMP unless thoroughly washed out.
- It is recommended to degrade HMP before cryopreservation. Wash the cell pellet at least twice before cryopreservation to remove residual dextranase.

Things to Note

When seeding cells with HMPs
Ensure cells and HMP are evenly distributed to maximize interaction.

When cell cluster appears
AimGel HMP and cells tend to aggregate in the well centre over time. Gently shake the culture plate to redistribute HMP and cells. **Avoid unnecessary pipette mixing in the first 4-5 days**, disturbing the HMP-cell clustering will cause suboptimal cell growth.

When to add new medium or split?
Monitor cell growth periodically by performing cell sampling and counting periodically. Supplement fresh medium or pass the cells to new culture vessels.

Other compatible media
We have tested AimGel Human NK Cell Activator compatibility in several commercial NK specific media. Adding AimGel HMPs in general boosted NK cell activation and growth over medium alone.

Website **LinkedIn** **Email**

AimGel
Part of our AimGel artificial cell line-up.

Allegrow

www.allegrowbiotech.com

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